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Automatic construction of large-scale IoT datasets with multi-strategy fusion

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Abstract With the advent of the "Internet+" era, the IoT has developed rapidly and is gradually penetrating into all fields of life[1]. While the scale of IoT devices is showing an explosive growth trend, the importance of IoT security is also becoming more and more prominent in the rapid development of the IoT[2]. In order to assist the identification of Internet-connected devices and further identify the vulnerability information of devices to achieve security protection for IoT devices, we construct a large-scale, diverse, and high-coverage dataset of IoT devices by automated and semi-automated means. In this dataset, each piece of data contains the information of the category, brand, and product model of the device. The large-scale, diverse, and high-coverage properties of the dataset are fully validated through our statistical analysis and experimental applications.

Keywords Large-scale Dataset, Web Crawler, Bi-LSTM+CRF, TF-IDF, Regular Expression

1 Introduction

Large-scale datasets are the premise for machines to learn knowledge and the basis for many downstream tasks. The construction of datasets usually includes the crawling of source data and the regularization of data. The technical problem we mainly solve in this paper is to design a data set construction system for downstream tasks such as device identification and IoT device knowledge graph construction, and to automatically collect the type, brand, product model and other information of IoT devices on a large scale[3]. The construction process of the system includes 6 parts, which are crawling of raw data, screening of IoT device categories, automatic identification of brand and model information, inspection and completion of brand and model information, Chinese-English alignment of categories and brands, and data deduplication. Among them, the automatic identification of brand and model information is implemented using two different methods (NER and Regular Expressions) (Figure 1).We also use two methods to verify and complete the brand and model information, the utilization of the URL page information of the product details page and the search engine + TF-IDF to obtain key information and a total of 384,875 pieces of data including 12 categories, 115 sub-categories, and 5,950 brands were obtained. This

#These authors contributed equally to this work.

article uses the python development language to start with a series of starting web pages using breadth-first, and automatically collects information in web pages in a tree structure. For some data irregularities, we use the regularization method pattern ='[A-Za-z0-9 \s\-\+ \. \(\)/]+') and named entity recognition based on Bi-LSTM+CRF method to process and extract the brand and model of each piece of data. Finally, we check and correct the data based on the structural information that may appear in the actual situation of each piece of data to complete the missing information.

2 Dataset Construction Methods

In this part, we will focus on the construction of original data, data classification and filtering, refinement of the data, and data standardization. For different problems in the data, we will use different methods to process and achieve the standardization and regularization of the dataset.

2.1 Construction of original data

The original data construction mainly starts from the regularized static websites such as industrial control network, security network, and product shopping guide network (Table 1), and adopts the crawler strategy of breadth-first search to capture the IoT device information (brand, model, product details page url, etc.)on these websites.

For the above website, we use the breadth-first[4] crawler to start from a series of starting web pages, extract all the sub-directory URLs in the web page, and put them into the queue for crawling in sequence.

In the specific implementation process, we first set up an empty crawler queue, sent the URL1 to be crawled into the crawler queue, and then read the URL1 in the queue in turn and collected all device types and corresponding URL2 in each URL1 page,and add the data to the queue in the format of <type, URL2>.Then we delete all URL1 in the queue, and read the <type, URL2> in the queue in turn, and get all the trademarks and brands in each URL2 page and the corresponding URL3,also add data to the queue in the format <type, brand, URL3>.In this process, if the brand information is "other", then add data to the queue in the format of <type, "other", URL3> and delete all <type, URL2> in the queue.

Finally, read the <type, brand, URL3> in the queue one by one, if there is model data in the current URL3 page, directly store the device model and the corresponding URL4, namely <type, brand, model, URL4>;If the model does not exist in the current URL3 page, then obtain the model through URL4 to the details page of the device, and store it in the data format of <type, brand, model, URL4>;If the model information cannot be obtained from the details page of the device, store it in the data format of <type, brand, "NULL", URL4> and delete all <type, brand, URL3> in the queue, so that the construction of the original data set has been completed[5].

2.2 Data classification and filtering

For the captured raw data, the device types (as shown in Tables 2 and 3) and the data in the IoT range were screened out to obtain 115 types (such as network cameras), which were semi-automatically processed in Chinese and English. The filtered data types are merged to obtain 12 relatively broad categories of IoT (such as video systems, each category contains several types), and semi-automatic processing in Chinese and English is carried out. The original data we collected may have a series of problems, such as the model field including product model information, brand information and other redundant content, and the lack of brand field or model field in some data. Therefore, it is

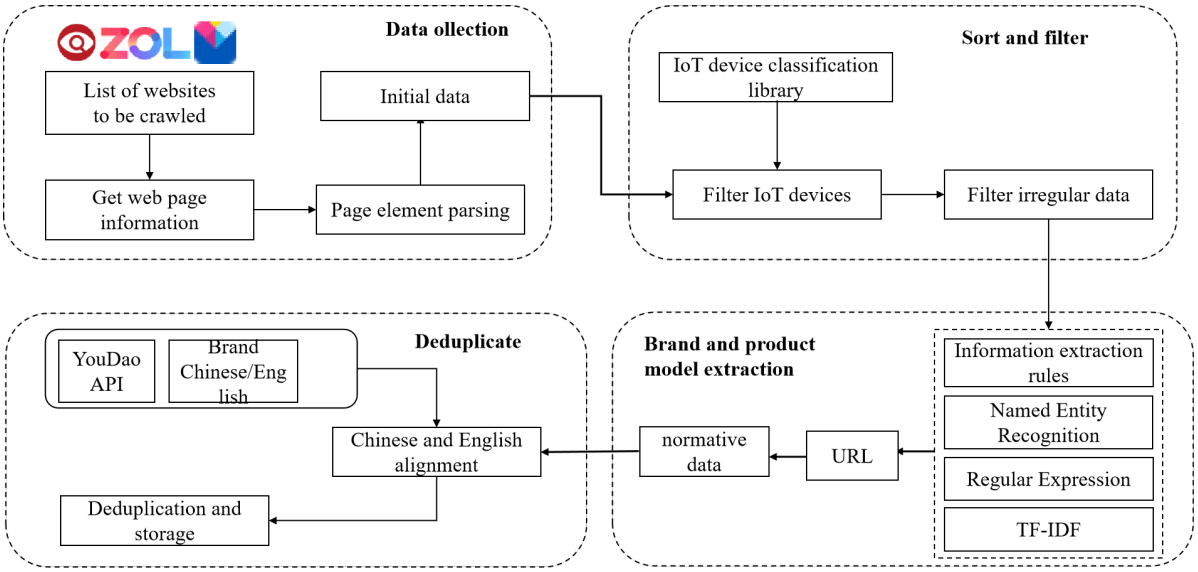

Fig.1. Overall architecture of dataset construction

Table 1. Introduction to the categories of IoT devices

Website Name	URL	Abbreviation	Data Volume
Zhongguancun Online Network	http://detail.zol.com.cn/subcategory.html	zol	508521
Pacific Internet	https://product.pconline.com.cn/category.html	pconline	197025
IT168	http://product.it168.com/category.html	it168	671938
Mobile Home Network	http://product.imobile.com.cn/phone.html	mobile	11496
Celestial net	http://product.yesky.com/category.shtml	yesky	261158
International Industrial Control Network	http://www.iianews.com/ca/sitemap.html	iianews	22808
Pacific Security Network	http://www.tpy888.cn	tpy888	338563
Huaqiang Wisdom Network	http://b2b.hqps.com/	hqps	41921
Security Knowledge Network	http://product.asmag.com.cn/	asmag	45930
China PLC Network	http://www.chinaplc.net/plcsupply/	plc	144875
China Security Industry Network	http://b2b.21csp.com.cn/	21csp	327144
Huaqiang Core City Network	http://www.hqchip.com/	hqchip	49316
Smart Home	http://www.znjj.tv/product/	znjj	40907
Hikvision official website	https://www.hikvision.com/cn/prnav_2.html	hikvision	2085
RFID World Network	http://www.rfidworld.com.cn/	rfidworld	3157

Table 2. Introduction to the categories of IoT devices

English abbreviation	English full name	Details
H_PersDev	Personal Device	Mobile phones, computers and other equipment used by major individual users
H_ServDev	Server Device	Main server and other equipment
M_RSS	Routing Switching System	There are mainly switches/gateways/routers and other equipment
M_CSS	Cyber Security System	Mainly refers to intrusion detection/firewall and other equipment
M_VSS	Video Surveillance System	Video surveillance equipment,etc.
M_ICS	Industrial Control System	Mainly refers to industrial control DCS/PLC and other equipment
M_OAS	Office Automation System	Mainly refers to equipment such as printers/scanners
M_VCS	Video Conference System	Mainly refers to video conference system equipment
M_HAS	Home Automation System	Mainly refers to smart TV/smart water heater and other equipment
M_SAD	Security Alarm Device	Alarm, alarm host, alarm system, etc.
M_BAS	Building Automation System	Control/elevator/access control equipment/air conditioning and ventilation monitoring system, etc.
M_OthCateDev	Other Category Device	Product category not listed

necessary to check the integrity and standardization of each data.

Table 3. Introduction of several types of M_VSS

English abbreviation	Category	English full name
IPCam	M_VSS	IP Camera
OCD	M_VSS	Other Camera Device
NVR	M_VSS	Network Hard Disk Video Recorder
DVR	M_VSS	Digital Video Recorder
VSD	M_VSS	Video Server Device
VOT	M_VSS	Video Optical Transceiver
NSD	M_VSS	Network Storage Device
VAR	M_VSS	Video Analyse Recoder
VMS	M_VSS	Video Management System
OMD	M_VSS	Other Monitor Device

2.2.1 Storage of complete data

For complete and pure data, first use these data to build a brand library, and then store it in json format after data standardization. Our dataset includes various IoT device information, and in Table 4 we show a part of the dataset for H_PersDev class.

Table 4. Partial data of H_PersDev class

Category	Type	Vendor	Product Model
H_PersDev	Computer	Think pad	tablet 236793gc
H_PersDev	Computer	Sony	Svt11227scw
H_PersDev	Computer	Dell	Latitude 2120
H_PersDev	Computer	Lenovo	ideapad s100-nfo
H_PersDev	Computer	Asus	fl5700u
H_PersDev	Computer	Samsung	nc110-p03

2.2.2 Processing of data redundancy

In our original data, if the brand field is redundant (more than 6 characters), the brand field is used as the input content. If the brand field is missing, input the content of the model field into the trained BiLSTM+CRF named entity recognition model to extract the brand information in the field. The model is obtained by training 16,000 pieces of data manually annotated (as shown in Table 5).

Table 5. Data annotation format

Labels	Full name	Representative entity
BV	Begin-Vendor	Start character of brand field
IV	Inside-Vendor	Brand field non-starting character
BP	Begin-Product	Start character of model field
IP	Inside-Product	Model field non-starting character
O	Other	Other characters

The model we used has 5 layers (as shown in Figure 2), which are input layer, embedding layer, BiLSTM[12][6] layer, CRF layer, and output layer from bottom to top.

The input layer is the first layer. In this layer, the brand field X that needs to be recognized by the name body is input into the model in the form of a single character.

The embedding layer is the second layer, which performs word embedding for each input sentence (that is, converts each input character into a vector), and its purpose is to convert logical characters into mathematical variables that the computer can recognize and calculate. In this model, we use randomly initialized word embeddings.

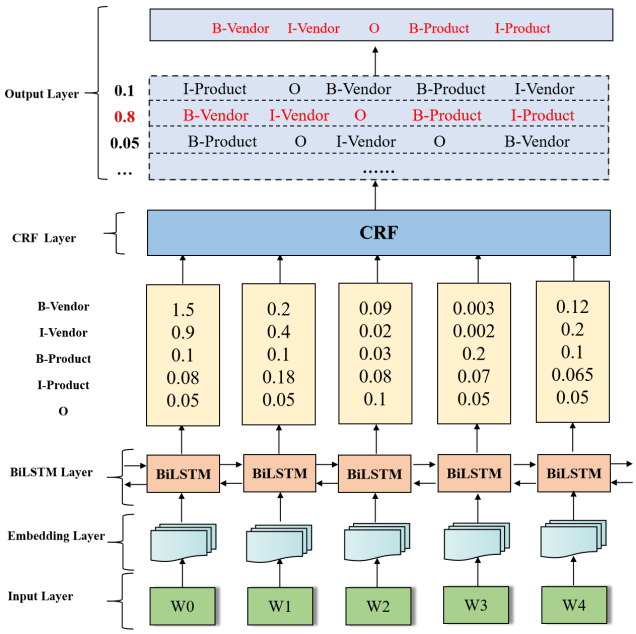

Fig.2. Brand and product model extraction models.

The Bi-LSTM layer is the third layer and its main purpose is to model text data[7]. At a certain point in the sequence, the forward-LSTM network is used to obtain the above information, and the backward-LSTM network is used to obtain the following information. By concatenating the output vectors of forward-LSTM and backward-LSTM, and the hidden layer output at the current moment is obtained through softmax calculation. Between the embedding layer and the Bi-LSTM layer, a Dropout parameter is also used. Dropout randomly invalidates some units in the forward calculation process to reduce the co-adaptation relationship between hidden units, which solves the overfitting problem and makes the neural network more robust to noisy data.

The CRF layer is the fourth layer, which takes the hidden layer output of Bi-LSTM as input and outputs all possible predicted label sequences Y. In the output layer, a (K+2)*(K+2) state transition matrix A is initialized during model training, and each element in the matrix represents the score (ie likelihood[8]) of the transition between labels. Here K represents the number of tags and 2 is the special tag for start and end. Equation (1) is used to calculate the likelihood of the predicted label sequence output by the CRF, and the probability of the true label sequence among all predicted label sequences is maximized (as in Equation (2)) as the objective function. During training, according to the principle of maximizing the objective function, A and the corresponding weights of the BiLSTM+CRF model are obtained by training.

$$s(X,y) = \sum_{i=0}^{n} A_{y_i,y_{i+1}} + \sum_{i=1}^{n} P_{i,y_i} \tag{1}$$

$$obj : max\left\{ p(y|X) = \frac{e^{s()}}{\sum_{\tilde{y}\in Y_X} e^{X,\tilde{y}}} \right\} \tag{2}$$

In the above formula, it represents all predicted label sequences output by CRF for a given input field X. For the predicted label sequence output by the CRF, the labels on the ith and i+1 characters represent the likelihood of being transferred from label to label; the output matrix of BiLSTM represents the likelihood score of the ith character being a label.

When applying the model, set the objective function formula (3) for the input field X, and output the optimal label sequence.We extract the corresponding character information (brand information) from specific tags (BV, IV)[9]

$$y^* = \underset{y\in Y_X}{argmax} \quad s(X,\tilde{y}) \tag{3}$$

2.3 Refinement of the data

Network search (Baidu) to obtain model information for each piece of data, and use the TF-IDF[10][11] algorithm to process the first 15 search results (Based on Baidu Baike corpus to find the most important 10 words in each result).We take the top 10 words with the highest frequency. If a word appears in the established brand library, the word will be used as the brand information, otherwise the word with the highest frequency will be used as the brand information. After doing a general statistics and analysis of the crawled data, it is found that the model field has the following characteristics: the model information of the device is a combination of letters, numbers and symbols, which is relatively regular and regular compared to other fields. So we use a regular expression (pattern ='[A-Za-z0-9 \s\-\+\.\(\)/]+')[12][13] to extract the model information in the model field. But if there are multiple model data in a sentence, take the longest one in the returned model list as the data model. At this point, the model information in the field is supplemented.

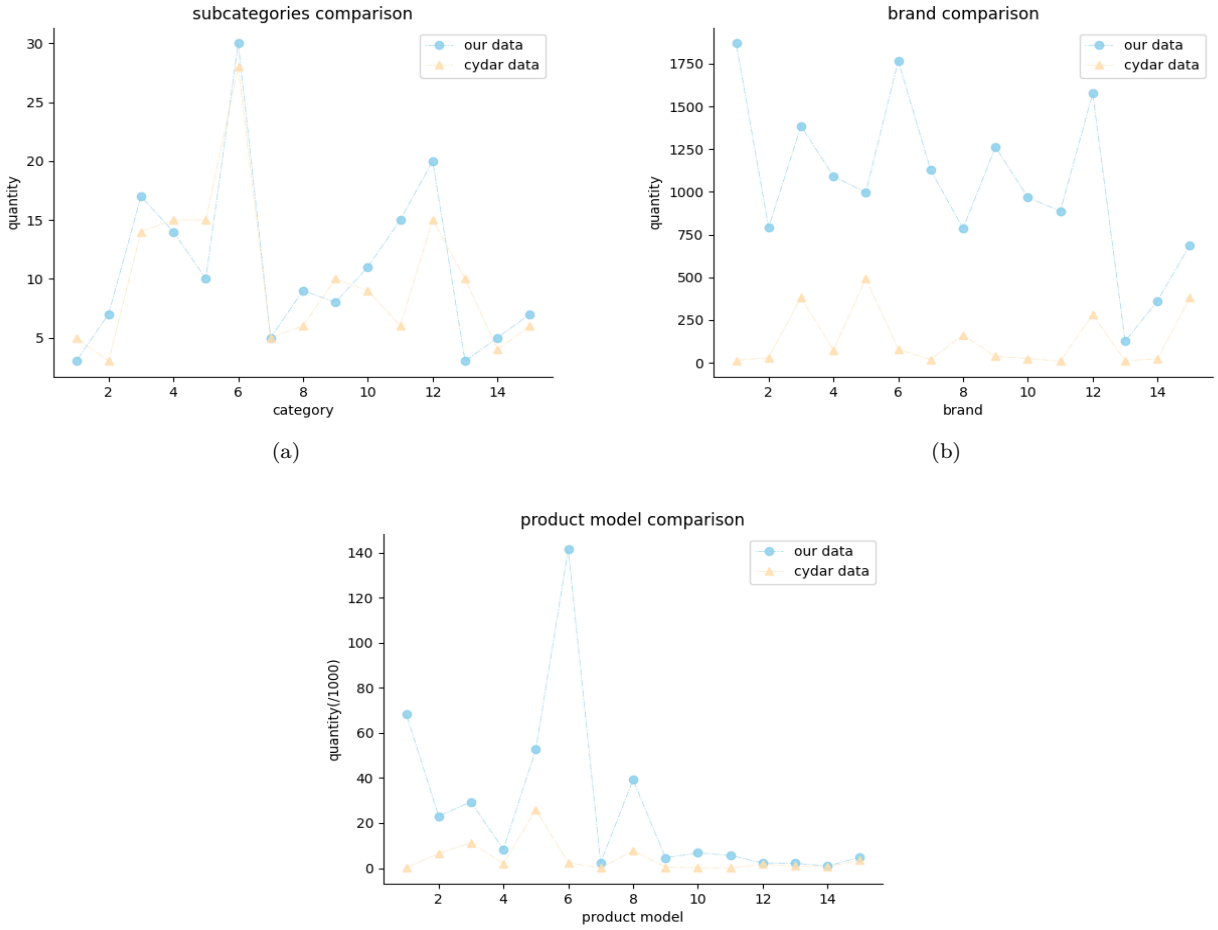

Fig.3. (a) subcategories comparsion (b) brand comparsion (c) product model comparsion

2.4 Data standardization

We check and correct brand and model information based on possible structured information based on the URL and detail page content of each piece of data. If the corresponding information cannot be extracted from the details page, the content of the obtained data will not be changed. In addition, the brand names of equipment captured by the website are confusing, for example: ”Hikvision”, ”Hikvision”, ”HIKVISION”, ”Hikvision” all refer to the same brand - Hikvision, therefore, it is necessary to convert all brands to a unified standard. Therefore, it is necessary to convert all brands to a unified standard.

With the help of Youdao dictionary API, we have realized the perfect mapping between Chinese and English in the brand library (if there are multiple translation results, we choose the one with the highest frequency), and replaced all device brands in the IoT device database with the corresponding lowercase English. For complete and purely data containing <type, brand, model, URL>, we store the dataset in the MongoDB database in json format.

3 Data Analysis and Application

After statistics, it is found that the dataset we constructed has a total of 15 categories of IoT device information and 164 subcategories(Table 6). In terms of

scale, it has reached the million level. In terms of diversity, from the number of brands in the table, we can see that each subcategory includes almost all brands in this category. In terms of coverage, as shown in the line chart in Figure 3, whether it is subcategory, brand or product model, the number of each type in our dataset is much higher than the number of types in the Cydar system(we will introduce in the application of our dataset). In addition, we did more than 1000 random lookup validations and found that only 3 random lookups resulted in results not included in our dataset. Finally, we apply our dataset to the Cydar system(as show in Figure 4), which is constructed through extensive screening of our dataset and some other datasets to quickly find relevant information about IoT devices. At the same time, we took the category of personal devices as an example to search and verify,we found that we have two more sub-categories than Cydar, and the number of brands is more than 1,000. In addition, we have more than 60,000 product models(as show in Table 6) than the Cydar system (as shown in Figure 3 (a), (b), (c)). From the line chart in Figure 3, we can find that our data set is largely superior to the data in the Cydar system, and we can also find that our data set has many brands and product models that the Cydar system does not have.

In addition, the categories and sub-categories in the practical Cydar system are shown in Figure 5. After analysis we can find that there are fewer main categories and subcategories in Cydar system than ours, which verifies the diversity of our dataset. This is because the classification methods used by the two are different. The classification method of our dataset is more detailed, while the classification method of the Cydar system is more general. We queried the two brands Thor and Hisense (as show in Figure 6) in the Cydar system and found no results, which verifies the high coverage of our dataset.Then we queried the hikvision brand (Part of the results are shown in Figure 7), we found a total of 1918 product models, which is close to the product model of hikvision in our dataset.Since one IP corresponds to multiple product models, we collect all product models, while the Cydar system only collects one product model in each IP, which leads to a gap in quantity. After the log4j vulnerability appeared not long ago, we combined the vulnerability warning system with this dataset to quickly and accurately give the relevant IoT device information with this vulnerability, which greatly reduced the loss caused by the vulnerability.

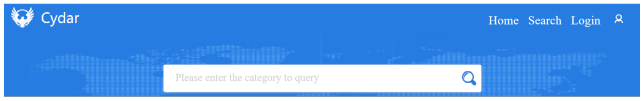

Fig.4. Home page of the Cydar system. [1]

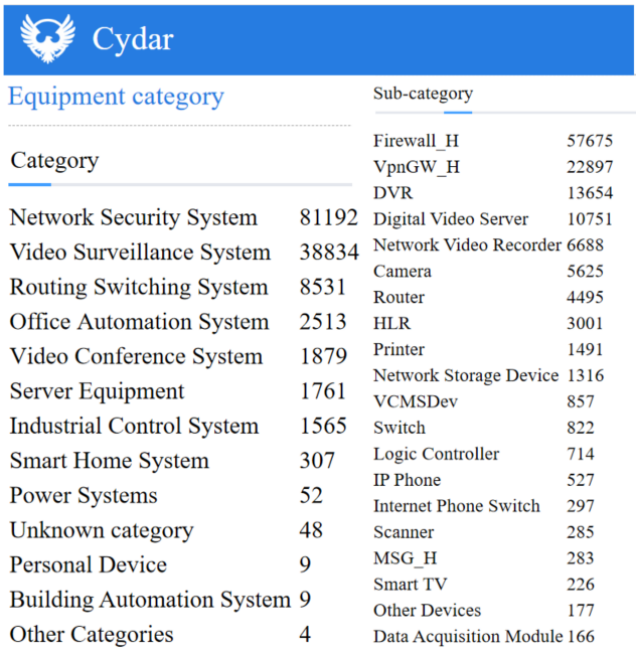

Fig.5. Categories and sub-categories in the Cydar system. [2]

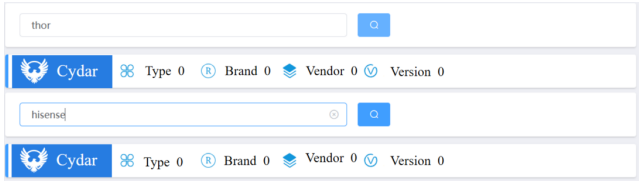

Fig.6. Search results for thor and hisense brands. [3]

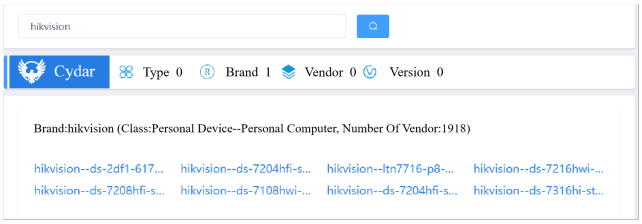

Fig.7. Search results for the hikvision brand. [4]

Table 6. Detailed number of main categories and subcategories

Main Categories	Subcate-gories	Number Of Brands	Product Model
Personal Device	3	1869	68402
Server Device	7	792	23077
Routing Switching System	17	1386	29378
Cyber Security System	14	1092	8265
Video Surveil lance System	10	997	52684
Industrial Control System	30	1766	141366
Building Automation System	5	1128	2451
Office Automation System	9	786	39260
Video Conference System	8	1262	4614
Home Automation System	11	968	6742
Electirc System	15	887	5662
Intelligent Healthcare System	20	1576	2316
Internet of Vehicles	3	126	2152
Security Alarm Device	5	362	974
Other Category Device	7	688	4802
Total NUmber	164	15685	392145

4 Conclusion

In order to assist in the identification of Internet-connected devices and further identify the vulnerability information of devices to achieve the security protection requirements for IoT devices, We design a dataset construction system and build a large-scale, diverse and high-coverage dataset based on this system, and through statistical analysis and application of the dataset, we demonstrate these three characteristics of the dataset. We hope that this dataset will play a driving role in future research on IoT device identification and IoT device security protection.

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